

**BOGUCHAR, Russian Federation**

WMO: 343360

Lat: 49.933N

Lon: 40.567E

Elev: 274

StdP: 14.55

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

**Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -9.4       | -3.3 | -15.0                         | 2.5 | -8.7 | -9.1 | 3.4 | -2.5 | 15.9                  | 27.9 | 13.9 | 27.8 | 3.2                      | 50   | 0.478 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.1                   | 94.0            | 68.6 | 90.2 | 67.7 | 86.9 | 66.6 | 71.3                | 86.9 | 69.9 | 85.3 | 68.5 | 82.7 | 5.6                  | 50   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.8                            | 99.5 | 76.3 | 65.1 | 93.8 | 75.0 | 63.5 | 88.7 | 74.3 | 35.3          | 86.4 | 34.0 | 85.4 | 32.9 | 82.9 | 82.4              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |  |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|--|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |  |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |  |
| 14.4              | 12.3 | 11.0 | DB                         | -15.3 | 97.8               | 8.0 | 3.8                                                | -21.0 | 100.5      | -25.7 | 102.8      | -30.1 | 105.0      | -35.9 | 107.7 |  |
|                   |      |      | WB                         | -15.6 | 75.7               | 7.9 | 3.5                                                | -21.3 | 78.2       | -25.9 | 80.3       | -30.3 | 82.3       | -36.0 | 84.9  |  |

**Monthly Climatic Design Conditions**

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 47.6  | 21.5  | 22.9  | 33.6  | 49.8  | 61.6  | 68.8  | 72.7  | 70.6  | 59.5  | 47.0  | 35.1  | 25.8  | (6)  |
| (7)  |                                                                                  | DBStd       | 20.49 | 11.97 | 12.17 | 8.55  | 7.87  | 8.33  | 7.05  | 6.13  | 6.99  | 7.70  | 8.47  | 9.45  | 11.11 | (7)  |
| (8)  |                                                                                  | HDD50       | 3635  | 884   | 758   | 510   | 99    | 10    | 0     | 0     | 0     | 12    | 161   | 450   | 751   | (8)  |
| (9)  |                                                                                  | HDD65       | 7067  | 1349  | 1178  | 974   | 457   | 167   | 40    | 7     | 26    | 197   | 560   | 896   | 1216  | (9)  |
| (10) |                                                                                  | CDD50       | 2741  | 0     | 0     | 1     | 93    | 370   | 566   | 705   | 638   | 297   | 67    | 4     | 0     | (10) |
| (11) |                                                                                  | CDD65       | 698   | 0     | 0     | 0     | 1     | 62    | 156   | 247   | 199   | 32    | 1     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH74       | 7612  | 0     | 0     | 0     | 29    | 693   | 1634  | 2617  | 2258  | 368   | 13    | 0     | 0     | (12) |
| (13) |                                                                                  | CDH80       | 3045  | 0     | 0     | 0     | 2     | 210   | 627   | 1132  | 984   | 88    | 2     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 4.7   | 4.9   | 5.2   | 5.6   | 5.2   | 4.6   | 4.4   | 4.0   | 3.9   | 4.0   | 4.4   | 4.9   | 5.3   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 20.2  | 1.8   | 1.3   | 1.2   | 1.3   | 1.8   | 2.3   | 2.2   | 1.5   | 1.6   | 1.4   | 1.9   | 2.0   | (15) |
| (16) |                                                                                  | PrecMax     | 28.2  | 4.1   | 3.1   | 4.7   | 4.3   | 5.2   | 6.4   | 7.1   | 4.7   | 6.3   | 4.8   | 4.4   | 4.4   | (16) |
| (17) |                                                                                  | PrecMin     | 12.7  | 0.2   | 0.0   | 0.1   | 0.1   | 0.2   | 0.4   | 0.2   | 0.0   | 0.0   | 0.1   | 0.0   | 0.2   | (17) |
| (18) |                                                                                  | PrecStd     | 4.4   | 1.0   | 0.7   | 0.8   | 0.9   | 1.3   | 1.5   | 1.3   | 1.0   | 1.4   | 1.0   | 1.1   | 1.1   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 44.4  | 46.4  | 60.6  | 77.7  | 89.9  | 94.3  | 99.6  | 99.5  | 87.2  | 74.4  | 58.6  | 47.6  | (19) |
| (20) |                                                                                  |             | MCWB  | 42.5  | 42.9  | 48.1  | 58.0  | 66.3  | 67.8  | 71.0  | 70.2  | 63.7  | 59.6  | 53.4  | 45.5  | (20) |
| (21) |                                                                                  | 2%          | DB    | 39.9  | 41.6  | 54.6  | 72.6  | 85.7  | 90.3  | 94.3  | 94.0  | 82.2  | 68.3  | 54.4  | 44.3  | (21) |
| (22) |                                                                                  |             | MCWB  | 38.2  | 39.2  | 44.7  | 55.9  | 64.7  | 67.7  | 69.4  | 68.0  | 62.2  | 57.1  | 50.8  | 42.5  | (22) |
| (23) |                                                                                  | 5%          | DB    | 37.0  | 38.4  | 49.5  | 68.4  | 81.6  | 86.9  | 90.2  | 89.4  | 78.3  | 63.7  | 50.8  | 41.2  | (23) |
| (24) |                                                                                  |             | MCWB  | 35.6  | 36.4  | 43.0  | 53.5  | 62.8  | 66.5  | 68.6  | 67.1  | 61.4  | 55.6  | 47.8  | 39.6  | (24) |
| (25) |                                                                                  | 10%         | DB    | 35.0  | 36.2  | 45.0  | 64.4  | 77.4  | 83.5  | 86.2  | 85.4  | 74.0  | 59.6  | 47.1  | 38.1  | (25) |
| (26) |                                                                                  |             | MCWB  | 33.8  | 34.6  | 40.2  | 52.1  | 61.3  | 65.7  | 67.3  | 66.2  | 59.8  | 53.9  | 44.6  | 36.7  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 42.9  | 43.8  | 50.1  | 60.4  | 69.7  | 72.8  | 75.2  | 72.5  | 68.1  | 62.9  | 54.1  | 45.9  | (27) |
| (28) |                                                                                  |             | MCDWB | 44.0  | 45.8  | 56.7  | 73.9  | 86.4  | 86.9  | 91.0  | 90.8  | 79.0  | 70.7  | 57.0  | 47.0  | (28) |
| (29) |                                                                                  | 2%          | WB    | 38.6  | 39.1  | 46.3  | 57.4  | 66.6  | 70.7  | 71.6  | 70.4  | 65.1  | 59.3  | 51.1  | 42.6  | (29) |
| (30) |                                                                                  |             | MCDWB | 39.6  | 40.8  | 53.0  | 68.3  | 81.1  | 84.2  | 86.9  | 87.8  | 76.2  | 64.7  | 54.1  | 44.2  | (30) |
| (31) |                                                                                  | 5%          | WB    | 35.8  | 36.8  | 43.2  | 55.4  | 64.6  | 68.8  | 70.1  | 68.9  | 63.1  | 56.6  | 48.1  | 39.7  | (31) |
| (32) |                                                                                  |             | MCDWB | 36.9  | 38.1  | 48.6  | 65.8  | 77.8  | 81.8  | 85.4  | 85.0  | 73.9  | 62.2  | 50.2  | 41.0  | (32) |
| (33) |                                                                                  | 10%         | WB    | 34.0  | 34.9  | 40.6  | 53.3  | 62.6  | 67.0  | 68.7  | 67.3  | 61.0  | 54.0  | 44.8  | 36.9  | (33) |
| (34) |                                                                                  |             | MCDWB | 34.8  | 36.0  | 44.8  | 62.6  | 74.9  | 79.9  | 83.0  | 81.8  | 71.3  | 59.2  | 46.8  | 38.0  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 9.9   | 11.6  | 13.6  | 19.5  | 21.4  | 20.9  | 21.1  | 23.4  | 21.1  | 15.4  | 10.1  | 8.9   | (35) |
| (36) |                                                                                  |             | MCDBR | 7.2   | 9.6   | 20.8  | 26.9  | 26.6  | 25.7  | 26.8  | 28.8  | 28.0  | 22.8  | 14.0  | 8.1   | (36) |
| (37) |                                                                                  |             | MCWBR | 6.7   | 8.1   | 13.0  | 13.7  | 11.6  | 10.1  | 9.7   | 10.4  | 12.6  | 13.2  | 10.4  | 7.8   | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 6.9   | 8.9   | 18.3  | 22.5  | 23.4  | 22.1  | 22.7  | 24.8  | 23.1  | 18.8  | 12.3  | 7.9   | (38) |
| (39) |                                                                                  |             | MCWBR | 6.9   | 8.0   | 12.5  | 12.4  | 11.8  | 10.2  | 9.8   | 10.5  | 11.9  | 12.2  | 10.1  | 7.8   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |       | 0.320 | 0.355 | 0.382 | 0.448 | 0.413 | 0.404 | 0.431 | 0.438 | 0.401 | 0.371 | 0.347 | 0.322 | (40) |
| (41) |                                                                                  | taud        |       | 2.108 | 2.020 | 2.095 | 2.072 | 2.271 | 2.327 | 2.255 | 2.217 | 2.292 | 2.324 | 2.279 | 2.184 | (41) |
| (42) |                                                                                  | Ebn at Noon |       | 215   | 237   | 256   | 252   | 268   | 272   | 261   | 252   | 249   | 235   | 209   | 197   | (42) |
| (43) |                                                                                  | Edh at Noon |       | 26    | 37    | 41    | 47    | 41    | 39    | 41    | 41    | 34    | 28    | 23    | 22    | (43) |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      |       | 329   | 641   | 1033  | 1387  | 1848  | 1976  | 1926  | 1732  | 1183  | 647   | 294   | 194   | (44) |
| (45) |                                                                                  | RadStd      |       | 68    | 117   | 161   | 143   | 162   | 165   | 128   | 120   | 136   | 94    | 61    | 37    | (45) |

**Historical Trends**

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   | (46) |
|      |                        | +1.12 | N/A     | N/A    | N/A     | N/A   | N/A   | -187        | -315  | N/A   | N/A   |      |
| (47) | Regional (6 neighbors) | +1.06 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -268  | N/A   | N/A   | (47) |

Nomenclature: See separate page